JACOB JIBU

Electrical & Electronics Engineer

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SUMMARY

Electrical and Electronics Engineering undergraduate with hands-on experience across power systems, electrical machines and drives, power electronics, and industrial automation, gained through an industrial internship in a manufacturing environment and applied project work. Comfortable with circuit design, hardware validation, and closed-loop control system development, using embedded platforms and programming as tools to implement and test electrical control logic rather than as the primary focus. Experienced leading multidisciplinary engineering teams from concept through testing. Seeking opportunities in electrical engineering, power systems, control systems, and industrial automation.

EDUCATION

Vellore Institute of Technology, Chennai, TN *08/2023 – Present*
B.Tech in Electrical and Electronics Engineering (7th Semester) — Cumulative GPA: 8.19 / 10
Profile Focus: Core electrical engineering coursework spanning power systems, electrical machines, power electronics, control systems, and circuit design.

MTHSS, Kaloor, Ernakulam, Kerala *07/2021 – 03/2023*
Higher Secondary Education — Overall Marks: 94.66%
Coursework included a focus on Electrical Systems.

EXPERIENCE

Automation & Industrial Intern | Apollo Tyres Ltd. *Jun 2025 – Jul 2025*
- Gained hands-on exposure to industrial automation, PLC-controlled manufacturing systems, and electrical distribution networks in a high-volume tyre manufacturing plant.
- Studied induction motors, electrical drives, industrial control panels, and preventive maintenance procedures on production-line equipment.
- Observed electrical equipment troubleshooting, industrial safety standards, and manufacturing operations across multiple production units.
- Analyzed production workflows, process automation, and manufacturing efficiency in an industrial electrical environment.

Electrical Systems Engineer, Recovery Systems | Team IGNITION *Aug 2025 – May 2026*
- Led electrical design and development of a controllable, paraglider-based rocket recovery system, targeting operation near 1 km apogee.
- Designed the power and actuation circuitry for servo-based recovery deployment, including neutral-position correction and stall-protection logic to prevent actuator damage.
- Engineered a wireless control link with transmitter/receiver acknowledgement and heartbeat monitoring to continuously verify link status between ground and flight hardware.
- Designed fail-safe electrical behavior, deployment/recovery mechanism logic, and link-loss disarm logic to ensure safe recovery under fault or communication-loss conditions.
- Coordinated hardware development, subsystem integration, testing, and reliability analysis across the recovery system team; managed project timelines and allocated responsibilities.

PROJECTS

Adaptive Matrix LED Headlamp System *Jan 2026 – Apr 2026*
- Designed PWM-driven LED driver circuitry and a multiplexed matrix drive for an adaptive automotive headlamp, reducing digital I/O usage while controlling individual LED zones.

- Developed zone-based adaptive dimming control logic for automotive power and lighting systems, using vehicle-detection input to adjust beam pattern in real time.

Technologies: *PWM • LED Driver Circuits • Automotive Power Systems • Arduino • Computer Vision (input signal)*

Smart Modular Climate Regulation Platform (SIH Internal Prototype) *Aug 2025 – Oct 2025*

- Developed a closed-loop climate control system for agricultural cold storage (onion storage), targeting conditions of approximately 2–4°C and 60–70% relative humidity.
- Designed the power and control scheme for a Peltier (thermoelectric) cooling module, exhaust ventilation fan, and pump, integrating DHT11 temperature and humidity sensors for feedback.
- Built sensor interfacing and actuator control firmware with remote monitoring for the closed-loop system.

Technologies: *Peltier Module • Control Systems • Temperature & Humidity Sensors • ESP32 • MQTT*

ESP32-Based Smart Energy Management & Predictive Load Shedding System *Jul 2025 – Dec 2025*

- Developed an energy management system for electrical load forecasting and demand-side control, applying machine learning to predict load patterns.
- Designed relay-based power switching circuitry for load shedding across multiple electrical load nodes.
- Implemented real-time monitoring and mobile-controlled load management with data acquisition across load nodes.

Technologies: *Relay Control • Load Forecasting • Power Distribution • ESP32 • MQTT*

Hack Computer Architecture System *Aug 2026 – Present*

- Building a 16-bit computer from basic logic gates based on the Hack architecture, including CPU (ALU, registers, program counter, control unit), RAM/ROM, and clock generation.
- Developing a video/display subsystem with SRAM-based video memory and VGA timing generation, including sync counters and pixel-count logic.

Technologies: *Digital Logic Design • Computer Architecture • VGA Timing*

SKILLS

Core Electrical & Electronics: Circuit Design, Power Electronics, Electrical Machines & Drives, Power Systems & Distribution, Control Systems, Industrial Automation, Hardware Validation, Signal Processing

Software & Design Tools: MATLAB, Simulink, LTspice, Keil µVision, Arduino IDE

Programming & Embedded Platforms: Embedded C, C, Python, Java, ESP32, ESP8266, Arduino

Additional Domains: IoT Systems, Wireless Communication, Sensor Interfacing, Machine Learning for Control Applications

Soft Skills: Teamwork, Problem Solving, Leadership, Communication, Adaptability

Languages: English (Full Professional Proficiency), Malayalam (Native), Tamil (Conversational), Hindi (Listening Proficiency)

EXTRACURRICULAR ACTIVITIES

Team IGNITION, VIT Chennai

- Participated in the design and development of experimental rocketry systems.
- Worked on recovery mechanisms, wireless communication systems, and electrical subsystem integration.

Smart India Hackathon (Internal Selection Round)

- Participated in the development of a closed-loop climate regulation system for agricultural applications.

Electrical & Electronics Projects

- Designed and developed multiple projects spanning power electronics, control systems, embedded systems and industrial automation.